FINITE PROJECTIVE MODULES AND FLATNESS

Abstract. Over any commutative ring, a module is finite projective if and only if it is finitely presented and flat. We give the constructive proof using finite relations, including the passage from one relation to a finite system and the construction of a linear map through a surjection. The definitions and constructions are those used in the formalisation.

1. Definitions and statement

Throughout, R is a commutative ring with identity, and all maps between modules are R-linear. The zero ring is allowed. All existence assertions mean mere existence: a witness may be used in a proof of a proposition, but its existence does not specify a chosen witness. A map $q : P \to M$ is surjective if, for each $x \in M$, there merely exists $v \in P$ such that $q(v) = x$.

For $n \geq 0$, let $e_1, \ldots, e_n$ denote the usual basis of R^n. We use the elementary identities

$$v = \sum_{i=1}^{n} v_i e_i, \qquad f\left(\sum_i a_i v_i\right) = \sum_i a_i f(v_i).$$

In particular, a linear map from R^n is determined by its values on the e_i. These identities also hold for $n = 0$, with empty sums.

Definition 1.1. A *finite presentation* of M consists of integers $n, m \geq 0$, a surjection $q : R^n \to M$, and vectors $a_1, \ldots, a_m \in R^n$ such that

$$q(a_j) = 0 \quad (1 \leq j \leq m), \qquad q(v) = 0 \implies \exists c_1, \ldots, c_m \in R \quad v = \sum_j c_j a_j.$$

The module M is *finitely presented* if such a presentation merely exists.

Writing $A : R^m \to R^n$ for the map whose columns are the a_j, this is precisely a finite exact presentation

$$R^m \xrightarrow{A} R^n \xrightarrow{q} M \longrightarrow 0.$$

The containment $\operatorname{im} A \subseteq \ker q$ follows from $q(a_j) = 0$ and linearity; the other containment is supplied explicitly.

Definition 1.2. The module M is *finite projective* if there merely exist an integer $k \geq 0$ and maps

$$p : R^k \longrightarrow M, \qquad s : M \longrightarrow R^k \quad \text{with} \quad ps = \mathrm{id}_M .$$

Thus finite projectivity is expressed as being a retract of a finite free module.

Definition 1.3. We use the single-relation criterion for *flatness*: for every $n \geq 0$, every $q : R^n \to M$, and every $a \in R^n$ with $q(a) = 0$, there merely exist an integer $k \geq 0$ and maps

$$B : R^n \longrightarrow R^k, \qquad u : R^k \longrightarrow M \quad \text{such that} \quad uB = q, \qquad B(a) = 0.$$

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To write this in terms of elements, put $x_i = q(e_i)$ and $y_l = u(e_l)$. If the entries of B are b_{li}, the criterion says that a relation $\sum_i a_i x_i = 0$ can be written as

$$x_i = \sum_l b_{li} y_l, \qquad \sum_i a_i b_{li} = 0 \quad \text{for every } l.$$

Thus the given relation follows from finitely many relations among scalars. Definition 1.3 is the formulation of flatness used in the formalisation.

Theorem 1.4. *An R-module is finite projective if and only if it is finitely presented and flat.*

2. One relation and finitely many relations

Lemma 2.1. *Let M be flat. Suppose $q : R^n \to M$ and $a_1, \dots, a_m \in R^n$ satisfy $q(a_j) = 0$ for every j. Then there merely exist $k \geq 0$ and maps*

$$B : R^n \longrightarrow R^k, \qquad u : R^k \longrightarrow M$$

such that

$$uB = q, \qquad B(a_j) = 0 \quad (1 \leq j \leq m).$$

Proof. Induct on m. For $m = 0$, take $k = n$, $B = \mathrm{id}_{R^n}$, and $u = q$.

For $m > 0$, first apply flatness to $q(a_m) = 0$. Obtain

$$B_0 : R^n \to R^{k_0}, \qquad u_0 : R^{k_0} \to M, \qquad u_0 B_0 = q, \qquad B_0(a_m) = 0.$$

For $j < m$ we have $u_0(B_0(a_j)) = q(a_j) = 0$. Apply the induction hypothesis to u_0 and these $m - 1$ vectors. It gives

$$C : R^{k_0} \to R^k, \qquad u : R^k \to M, \qquad uC = u_0, \qquad C(B_0(a_j)) = 0 \quad (j < m).$$

Set $B = CB_0$. Then $uB = uCB_0 = u_0 B_0 = q$. The first $m - 1$ vectors are killed by construction, and $B(a_m) = C(0) = 0$. □

The finite system in this lemma is obtained by induction from the single-relation definition; it is not an additional hypothesis on M.

3. Factoring through a surjection

Lemma 3.1. *Let $q : P \to M$ be surjective, and let $b : P \to N$ satisfy*

$$q(v) = 0 \implies b(v) = 0.$$

There is a unique linear map $s : M \to N$ such that $sq = b$.

Proof. If $q(v) = q(w)$, then $q(v - w) = 0$, so $b(v - w) = 0$ and $b(v) = b(w)$.

For each $x \in M$, there is therefore a unique $y \in N$ with the property

$$q(v) = x \implies b(v) = y \qquad (v \in P). \tag{1}$$

Existence follows by using one preimage of x; independence of that preimage is the equality just proved. Uniqueness follows because x has a preimage. Define $s(x)$ to be this unique value. This construction uses unique choice of a value, rather than a choice of a preimage for every x.

By (1), $s(q(v)) = b(v)$. To prove additivity, use preimages $q(v) = x$ and $q(w) = x'$ inside the proof of the equality. Then

$$s(x + x') = s(q(v + w)) = b(v + w) = b(v) + b(w) = s(x) + s(x').$$

Similarly, for $c \in R$,

$$s(cx) = s(q(cv)) = b(cv) = c\, b(v) = c\, s(x).$$

These are equalities in sets, so the mere existence of the finitely many preimages suffices to prove them. Hence s is linear. Finally, any map satisfying $sq = b$ must take every x to the unique value in (1), proving uniqueness. □

4. Finite presentation and flatness give a retract

Proof. Suppose M is finitely presented and flat. Since our conclusion is an existence assertion, we may use a finite presentation as in Definition 1.1. Lemma 2.1 applied to its relators gives k, $B : R^n \to R^k$, and $u : R^k \to M$ with

$$uB = q, \qquad B(a_j) = 0 \quad (1 \le j \le m).$$

The map B kills all of $\ker q$. Indeed, if $q(v) = 0$, the presentation gives $v = \sum_j c_j a_j$, and hence

$$B(v) = \sum_j c_j B(a_j) = 0.$$

By Lemma 3.1, B factors uniquely as

$$B = sq \qquad \text{for a linear } s : M \to R^k.$$

Consequently $usq = uB = q$. For $x \in M$, use any preimage $q(v) = x$ to obtain

$$us(x) = usq(v) = q(v) = x.$$

Thus $us = \mathrm{id}_M$, and (k, u, s) exhibits M as a retract of R^k. The required retract therefore merely exists. □

5. A retract is finitely presented and flat

Proof. Suppose M is finite projective. We may use maps

$$p : R^k \to M, \qquad s : M \to R^k, \qquad ps = \mathrm{id}_M .$$

We construct a finite presentation and verify the single-relation criterion directly.

For the presentation put

$$h = \mathrm{id}_{R^k} - sp, \qquad a_i = h(e_i) = e_i - s(p(e_i)) \quad (1 \le i \le k).$$

The map p is surjective, with preimage $s(x)$ for $x \in M$. Moreover,

$$p(a_i) = p(e_i) - psp(e_i) = 0.$$

If $p(v) = 0$, then $h(v) = v - s(0) = v$. Writing $v = \sum_i v_i e_i$ and using linearity gives

$$v = h(v) = \sum_i v_i h(e_i) = \sum_i v_i a_i.$$

Thus the a_i generate $\ker p$, and we have the finite presentation

$$R^k \xrightarrow{\mathrm{id} - sp} R^k \xrightarrow{p} M \longrightarrow 0.$$

To prove flatness, let $q : R^n \to M$ and let $a \in R^n$ satisfy $q(a) = 0$. Take

$$B = sq : R^n \to R^k, \qquad u = p : R^k \to M.$$

Then $uB = psq = q$ and $B(a) = s(q(a)) = s(0) = 0$. This is exactly Definition 1.3.

Both constructions are made from a retract. Mere existence of a retract gives mere existence of the presentation, and, for each relation separately, mere existence of its explanation. This proves finite presentation and flatness, completing the theorem. □

Correspondence with the formalisation

The proof above follows the four files below in `sag/chatelet`. All definitions are over an arbitrary commutative ring R with a set as its underlying type.

Mathematical statement	**File and declaration**
Finite linear combinations and their images under linear maps	`DescentModuleMaps.seg`: `Free.LinearMap.equation`, `Killed.theorem`
Flatness, Definition 1.3	`DescentFlat.seg`: `At.Flat`
Finite systems, Lemma 2.1	`DescentFlat.seg`: `At.OneToMany.theorem`
Factor through a surjection, Lemma 3.1	`DescentSurjectiveFactor.seg`: `At.factor`, `At.beta`, `At.linear`
Finite presentation and finite projectivity	`DescentProjective.seg`: `At.Presentation`, `At.Retract`, `At.FinitelyPresented`, `At.FiniteProjective`
Finitely presented and flat implies finite projective	`DescentProjective.seg`: `At.fp-flat-to-projective`
A finite projective module is finitely presented and flat	`DescentProjective.seg`: `At.projective-to-fp`, `At.projective-to-flat`
Theorem 1.4	`DescentProjective.seg`: `At.finite-projective-iff-fp-flat`

The main declaration is

```
DescentProjective.Over[R,setR].At[M].
    finite-projective-iff-fp-flat.
```

In `DescentFlat`, the data k, B, u with $uB = q$ are recorded in `FiniteFactor`; `Explanation` additionally records that B kills the specified relators. In `DescentProjective`, the reverse implication constructs the map s of Section 4, and the forward implication uses exactly the relators $(\mathrm{id} - sp)(e_i)$ of Section 5.